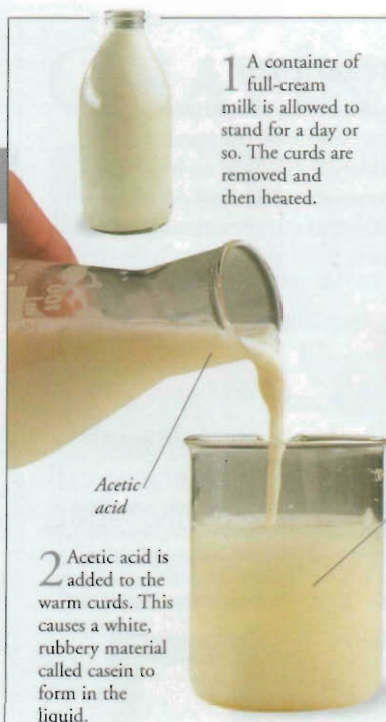


1 A container of full-cream milk is allowed to stand for a day or so. The curds are removed and then heated.



2 Acetic acid is added to the warm curds. This causes a white, rubbery material called casein to form in the liquid.

Acetic acid

Casein forms in curds.

3 The casein is removed by straining the liquid. It is kneaded in warm water and then dried.

Casein hardens as it dries.

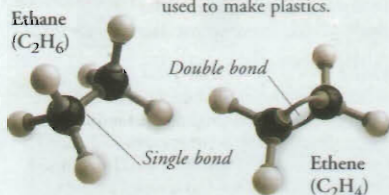
4 Casein can then be mixed with other materials to form paint. The casein is used to bind the pigment to the surface.

Casein-based paint

Aliphatics

Organic compounds containing a chain of carbon atoms linked by single, double, or triple bonds are called aliphatics. The aliphatic ethane occurs in natural gas, and ethene is used to make plastics.

Ethane (C_2H_6)



Double bond
Single bond

Ethene (C_2H_4)

Fats and oils

Liquid vegetable oil is an unsaturated fat – a type of fat in which some of the carbon atoms are linked by double bonds. When the oil reacts with hydrogen, the double bonds break and the carbon atoms link up with extra hydrogen atoms, forming a solid fat such as margarine. Solid fats contain only single bonds between their carbon atoms. These fats are said to be saturated because they cannot bond with more hydrogen atoms.



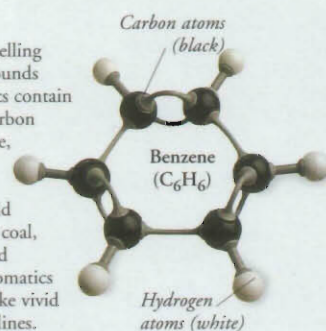
Liquid oil



Solid fat

Aromatics

The strong-smelling organic compounds called aromatics contain a ring of six carbon atoms. Benzene, the simplest aromatic, is a colourless liquid obtained from coal, natural gas, and petroleum. Aromatics are used to make vivid dyes called anilines.



Carbon atoms (black)

Benzene (C_6H_6)

Hydrogen atoms (white)

Polymers

Polymers are giant molecules that consist of winding chains of thousands of small organic molecules called monomers.

Fats, starches, and proteins are natural polymers; plastics and artificial fibres are made of synthetic polymers. Polythene contains polymers made of many ethene monomers joined together.

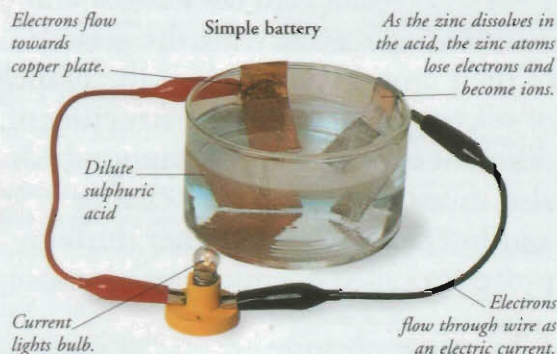
Plastics, such as polythene and PVC, are made up of polymers.



Plastic products

Electrochemistry

The study of the relationship between electricity and chemical substances is called electrochemistry. Many compounds consist of electrically charged particles called ions, which form when atoms lose or gain electrons. A battery uses a chemical reaction to generate an electric current.



Electrolysis

The process of splitting up a compound by passing an electric current through it is known as electrolysis. Two metal or carbon rods called electrodes are placed in the compound and connected to a battery. As electricity flows through the compound, positive ions are attracted to the negative electrode (the cathode), and negative ions are attracted to the positive electrode (the anode), causing the compound to split apart.



Research into respiration



Geochemist examining rocks

Biochemistry

The study of the chemistry of living organisms and the chemical processes, such as respiration, that take place within them is called biochemistry. The discoveries of biochemists are used in industry, medicine, and agriculture.

Geochemistry

The Earth's composition and the chemical structure of rocks are studied in geochemistry. The findings of geochemists give us a greater knowledge of the Earth's history and help us to find ores, minerals, and other resources.

Alfred Nobel

The Swedish chemist Alfred Nobel (1833–96) invented the explosives dynamite in 1867 and gelignite in 1875, which made him very rich. On his death, he left his vast wealth to pay for a series of annual awards – the Nobel Prizes – for achievements in science, art, and medicine.



Timeline

2 BC Egyptian alchemists try to change "base" metals, such as lead, into gold.

1661 Robert Boyle, an Irish scientist, realizes that chemical reactions can be explained by the existence of small particles.

1770s Antoine Lavoisier investigates compounds such as air and water.

1807 Englishman Humphry Davy uses electrolysis to discover the element sodium.



1808 English scientist John Dalton proposes that each element has its own unique type of atom.

1830s German chemists focus on studying carbon and its compounds.

Davy's equipment

1909 American Leo Baekeland makes the first fully synthetic plastic – Bakelite.



1939 Linus Pauling, an American chemist, explains the nature of chemical bonds between atoms and molecules.

Bakelite radio set

FIND OUT MORE

ACIDS AND ALKALIS

ATOMS AND MOLECULES

DYES AND PAINTS

ELECTRICITY

ELEMENTS

GLASS

MEDICINE

MIXTURES AND COMPOUNDS

PLASTICS AND RUBBER